

Post AI Era Datacenters Deepdive

Papadakis Konstantinos Fotios
AEM: 10371

Contents ≡

1. Status quo
2. Datacenters Pillars
3. Energy Requirements
4. Network topology
5. Environmental Impact
6. Health
7. Finance
8. Sociopolitical Aspect



Status Quo

What's the driving force of datacenter construction and expansions?

Artificial Intelligence

- There always were Datacenters
- Many ambitious datacenter projects
- Focus on new facilities and expansions post AI era
- Push towards AGI
- Technology replacing humans in most fields
- note-to-self: why does the AI crisis seem to be different than technology not creating that many new jobs?



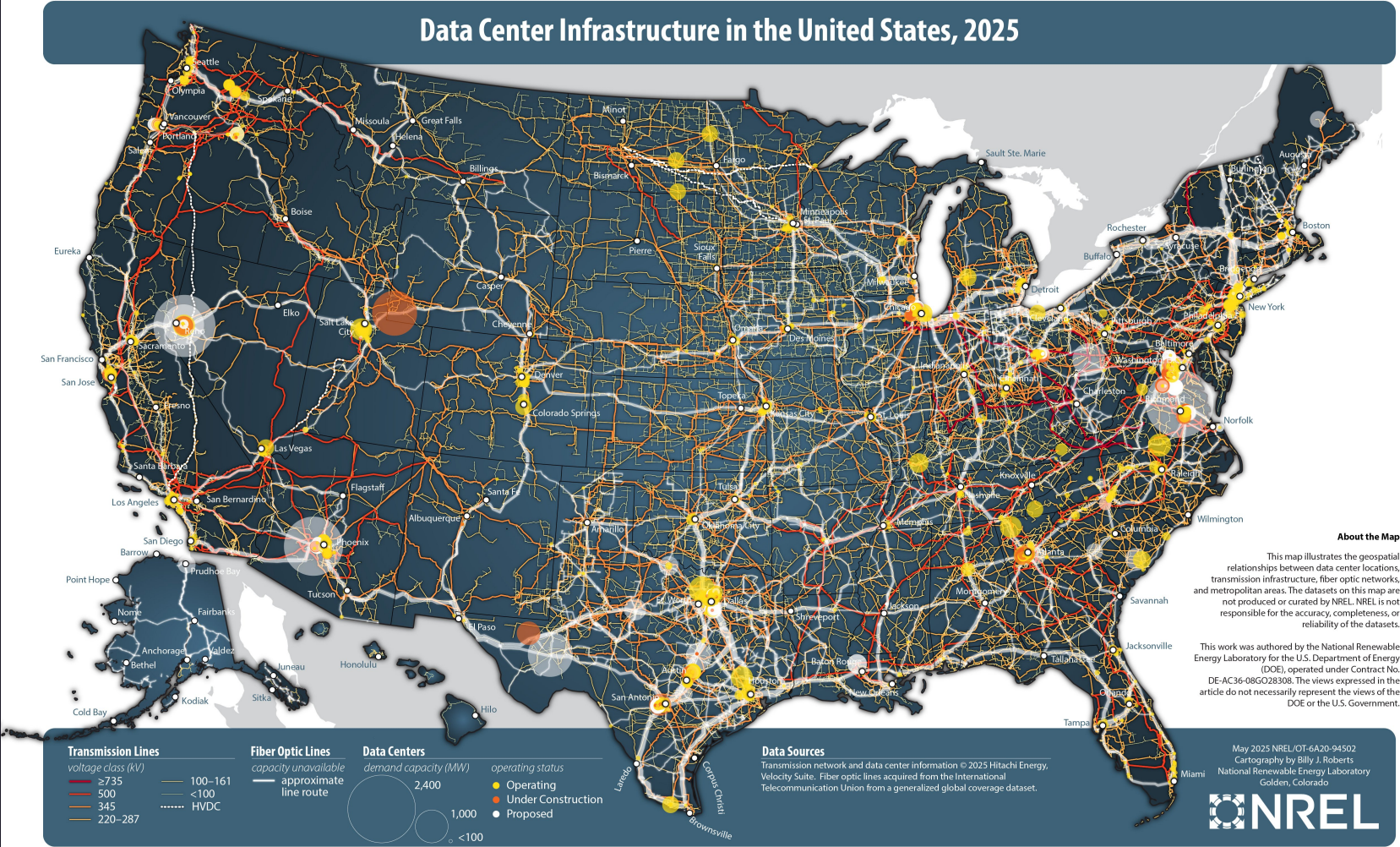
New Datacenters

The datacenters that are responsible for running and training the worlds most advanced LLMs.

Map

Most Activity

	State
1	Virginia
2	Texas
3	California
4	Illinois
5	Ohio



LLM Facilities

Datacenter	Location	Company	LLM
Canton Facility	Mississippi	Amazon	-
Monterey Park Data Center	California	California	-
Stargate Project	Saline Township, Michigan	OpenAI, Oracle	-
Colossus	Memphis	xAI	



Stargate

Specifications

Specs	Value
Investment	\$500 billion
Location	Abilene, Texas
Company	OpenAI & Oracle
IT Capacity	4.5 GW

based on ¹⁰

5 new sites

- Shackelford County, Texas
- Milam County, Texas
- Doña Ana County, New Mexico
- Lordstown, Ohio
- a mystery site located somewhere in America's Midwest.

End goal being 10GW of combined capacity.

Chips

- Nvidia GB200 racks



Monterey Park Datacenter



Specifications

Specs	Value
Investment	
Location	
Company	
IT Capacity	



Canton Amazon

Specifications

Specs	Value
Investment	\$10 billion
Location	Canton, Mississippi
Company	Amazon
IT Capacity	



Colossus

Specifications

Specs	Value
Investment	
Location	
Company	xAI
IT Capacity	

**~110,000 NVIDIA GB200 NVL72 GPUs
(targeting 1.1 PFlops FP8 compute).
usually small fp numbers needed for**

Energy Requirements ⚡

Ways to fulfil the insatiable hunger of datacenters for electricity.

Local Energy source candidates

City main line isn't enough. This necessitates the use of backup turbines power by:

- **Diesel**
- **Field gas**: Gas extracted from a production well before it enters a natural gas processing plant ³¹.
- **CNG**: Compressed Natural Gas, mainly methane compressed at a pressure of 200 to 248 bars ³².
- **LPG**: Liquefied Petroleum Gas, a mixture of propane and butane liquefied at 15 °C and a pressure of 1.7-7.5 bars ³².
- **LowBTU gases**: Natural gas, which at the wellhead has a gross heating value ≤ 450 BTU. Part of petroleum refining and crude oil, natural gas production.

Fuel Comparison

Comparison	CNG	LPG
Constituents	Methane	Propane, Butane
Energy density	Lower caloric value	Higher caloric value
Cost	Cheaper	More expensive
Emissions	Less than LPG	Less than gasoline
Safety	Disperses quickly	Settles to ground, but highly inflammable

CNG risk of ignition is low so even though LPG is highly inflammable it is the more likely of the two to ignite.

Diesel Engine Turbines

Monterey Park Data Center , Canton Mississippi

Methane Gas Turbines 🛢️

Colossus specifications

- natural gas turbines for primary generation
- batteries for stability
- grid for long-term scalability

Elon Musk's xAI data center Methane gas turbines, Colossus. ²⁴

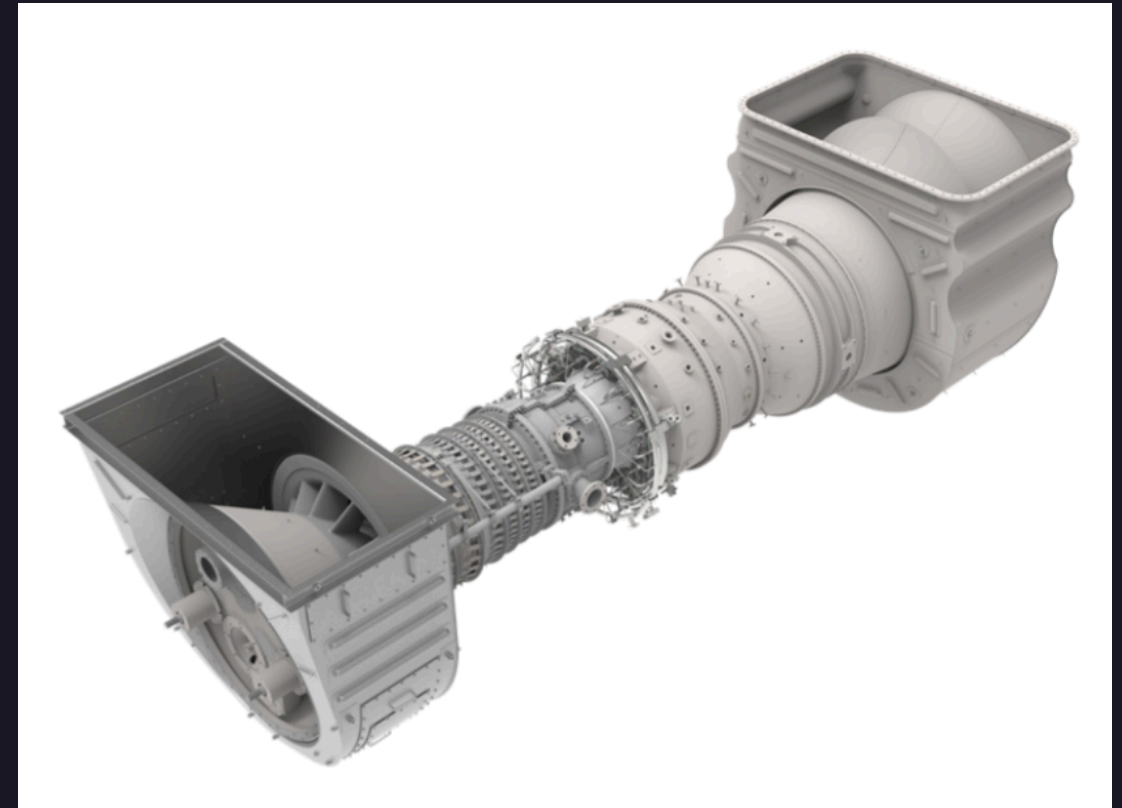


- 7 Solar Turbines Titan-350 ²²
- 12 Solar Turbines SMT-130 ²³

Solar Turbines: a Caterpillar subsidiary

Solar Turbines Titan 350

Specifications	Value
Power output	35-39MW
Thermal efficiency	~40%
Fuel types	Natural gas, Propane, Low BTU gases



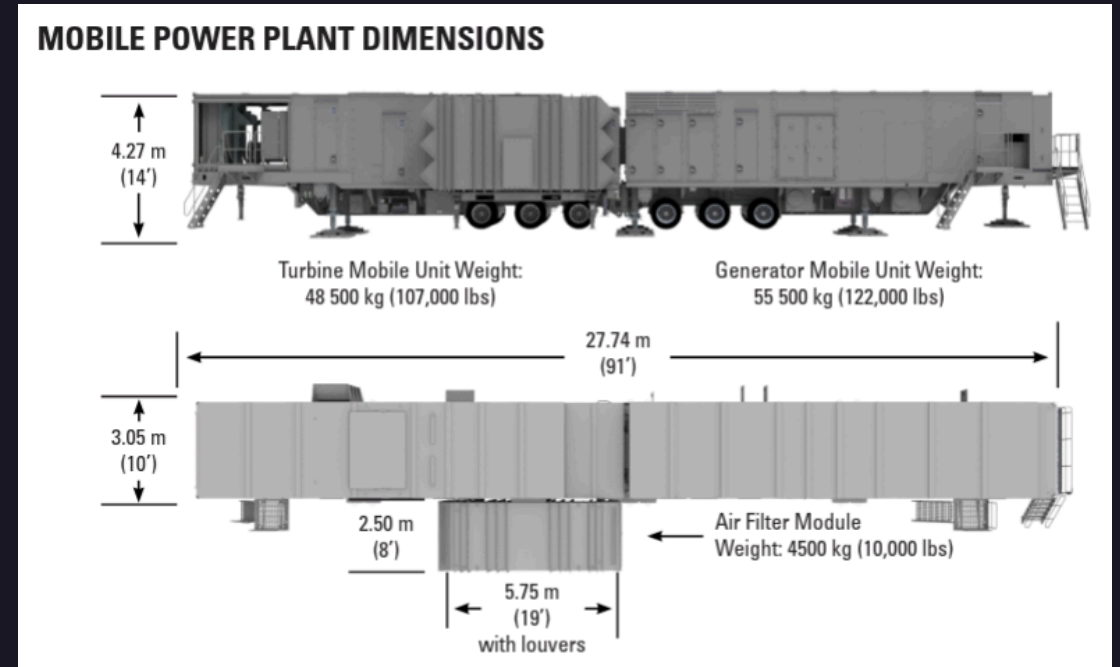
source^{22, 24, 27}

Solar Turbines Titan 350

Advanced Specifications	Value
Heat Rate	8845 kJ/kW-hr - 8780 kJ/kW-hr
Exhaust Flow	371980 kg/hr - 387820 kg/hr
Exhaust Temp	460°C - 490°C
Emissions	25 PPM NOx

Solar Turbines SMT 130

Specification	Value
Power output	16MW
Thermal eff.	~36%
Fuel types	Field Gas, CNG, LPG
Heat Rate	10160 kJ/kWe-hr
Exhaust Flow	202510 kg/hr
Exhaust T	490°C
Emissions	25 PPM NOx



Fully-Integrated Mobile Power Plant
Powered by Titan 130 ^{23, 24, 28}

+ Added Diesel fuel support

SoloNOx

SoloNOx is the technology enabling Solar Turbines to reduce NOx and CO emissions.³⁰

Offers a robust:

- 9ppm NOx,
- 15ppm CO, and
- 15 ppm UHC

emissions warranty for natural gas fuel.

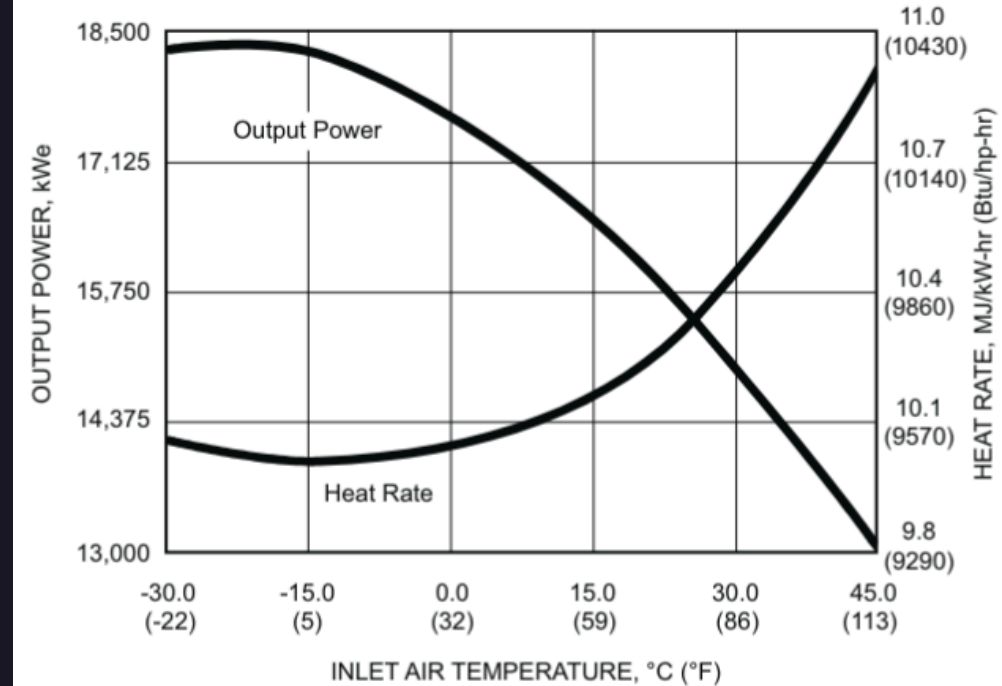
Available power

SMT 130 supports Diesel fuel boosting for added flexibility.

Titan 130 available power graph

Colossus power generation around 1.1 GW by Q2 2027

Available Power*



* SoLoNOx

Tesla Megapacks

Megapacks are large scale energy storage. *Many many Batteries*

Used for:

- Power Backup
- Resilience for outages
- Demand-response
- Turbine ramping

They can provide 4 hours of power. They are planing to deploy 200 megapack for about 1 gigawatt hour of buffering. Long term solar powered facility but near-term it is Megapacks and gas turbines that is being used as a bridge to the future.

Sustainable Pledges

Network topology

Network topology optimization

How are the cables routed

Environmental Impact

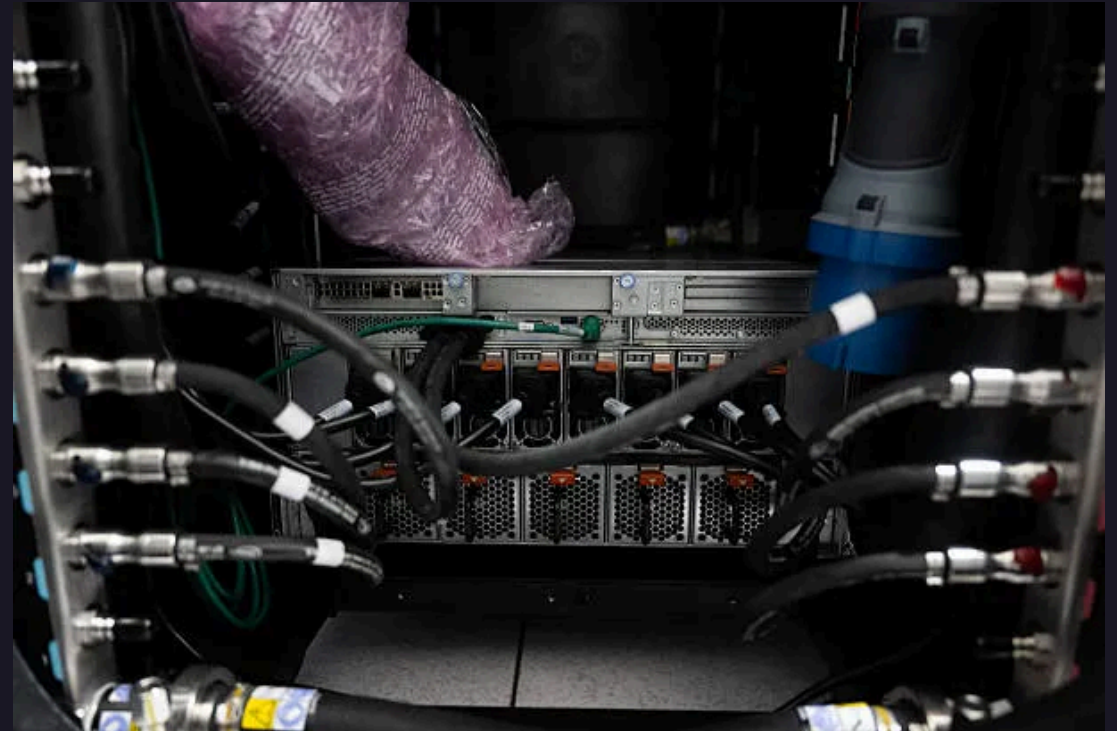
Water Shortage 💧

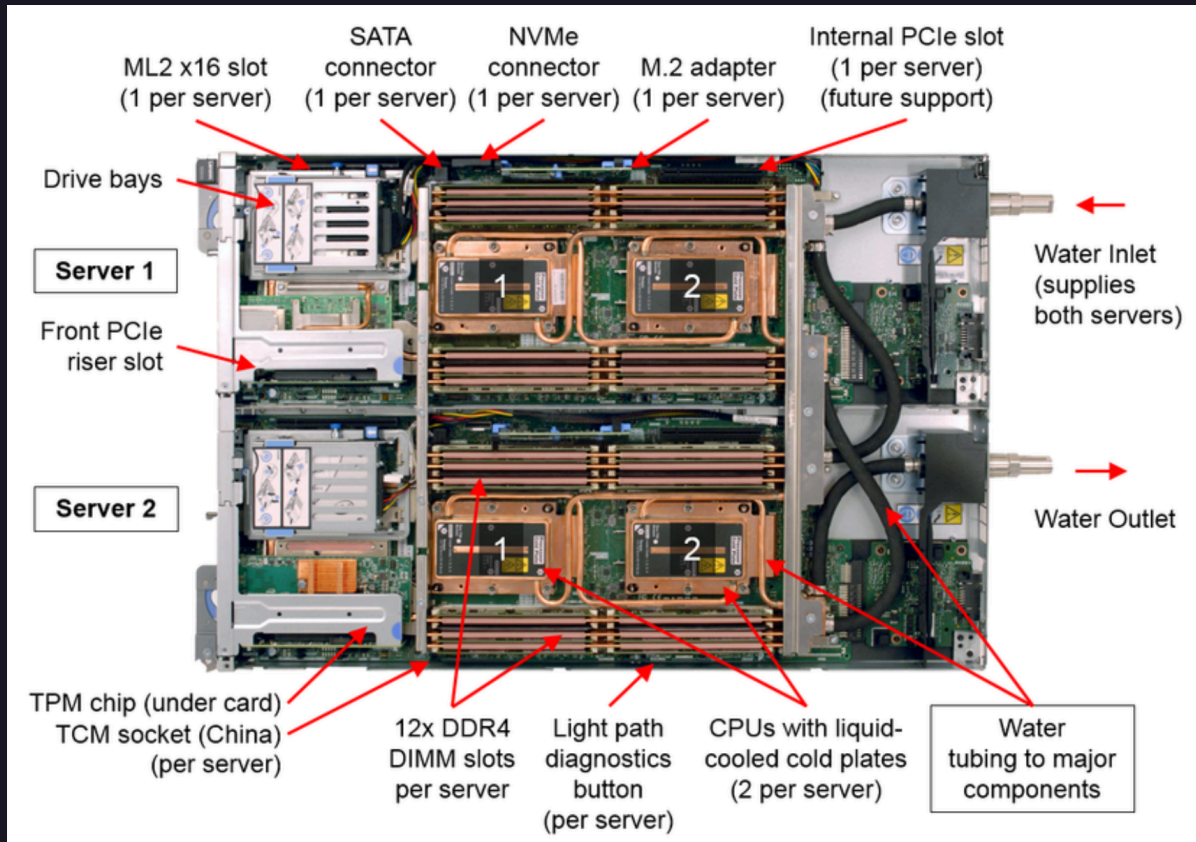
**Servers are usually water cooled.
Most of the water turns into steam and
is released into the atmosphere**

Alternatively it is

**Water is chosen because it is more
thermally conductive than air**

- Air: $0.024\text{W/m}^2\text{K}$
- Water: $0.58\text{W/m}^2\text{K}$



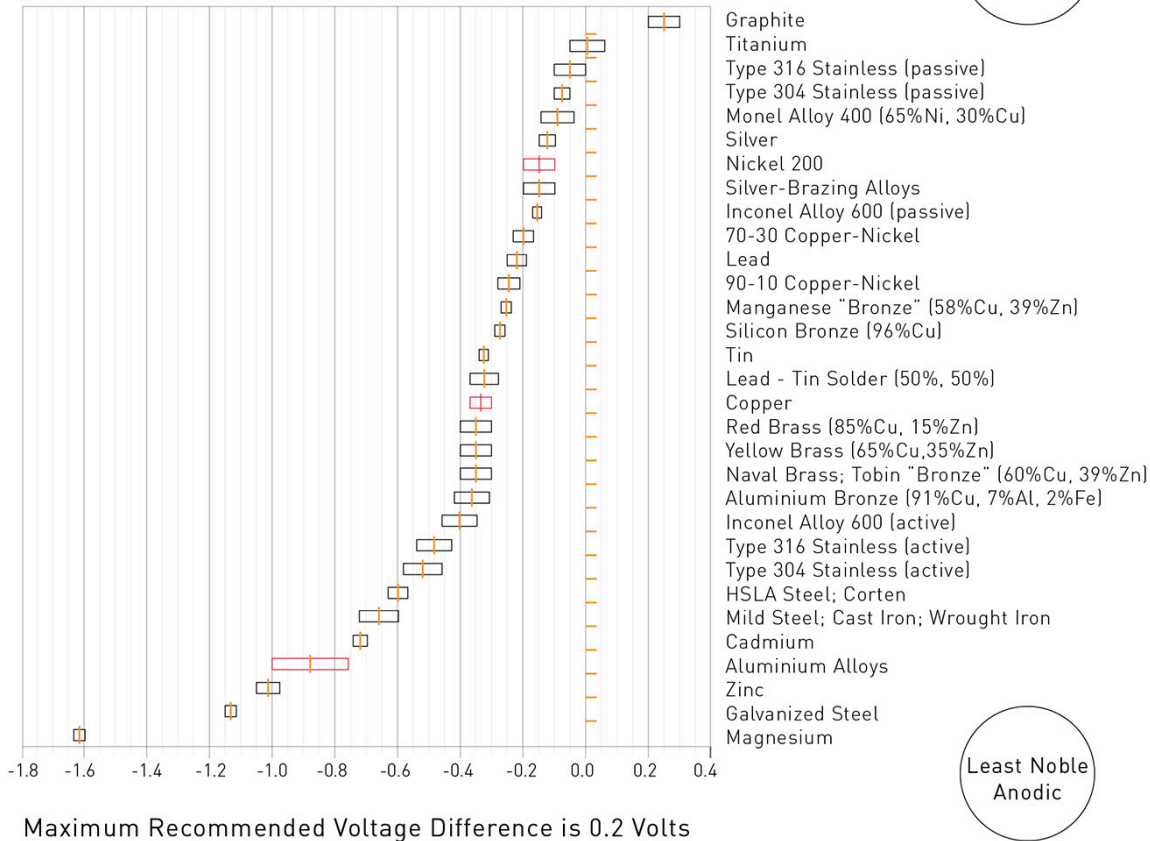


Watercooled server

- Copper piping
-  **Great thermal conductivity**³³
 - Copper: $401\text{W/m}^2\text{K}$
 - Aluminum: $205\text{W/m}^2\text{K}$
-  **More expensive per tonne**³⁴
 - Copper: $12808\$/\text{t}$
 - Aluminum: $3385\$/\text{t}$

The Galvanic Series Chart

Average Voltage in Seawater



Galvanic Corrosion

Galvanic is an electrochemical process that occurs when two metals with different electrochemical activity are in contact with each other (like copper and aluminum).

The more noble, passive metal (copper or nickel) drives the corrosion of the active, less noble metal (aluminum), where the passive metal remains fairly unharmed. ³³

Energy sources and emissions

Electricity-generating turbines were exempt from requirements for air quality permits. ¹⁴

loophole allowing the operation of generators without permits so long as the machines did not sit in one place for more than 364

xAI eventually received permits for 15 turbines at Colossus 1 and is now operating 12 permitted machines at the site.

- Net annual nitrogen oxide emission reductions of up to 296 tons by 2032," ⁴
- diesel generators spike local NOx levels

Health Effects

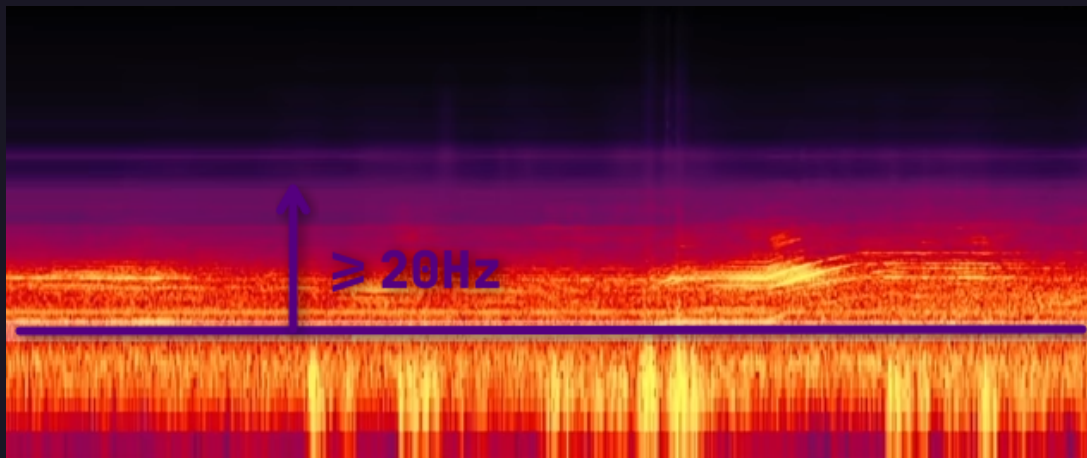
Canton

Residents reported:

- Lung irritation
- Breathing difficulties
- Construction dust that settled over the area
- Elevated childhood asthma rates ⁶
- Cooling towers pull millions of gallons of water daily from the already-stressed Big Black River system,
- while weekly tests of backup diesel generators spike local NOx levels and worsen the area's elevated childhood asthma rates. ⁶

Infrasound

Infrasound is low frequency sound below the lower limit of human hearing (around 20Hz) ¹



: loud, Y-axis: frequencies, X-axis: time

Known Health Effects ¹³

- Spike in cortisol levels (stress, hypertension) ^{14, 15}
- Nausea, Dizziness ¹²
- Vibroacoustic disease ¹⁶
- High frequency hearing loss ¹⁷
- Shortness of breath ¹⁸
- Anxiety, Depression ¹⁹

Sustained audible noise pollution

Constant sound exposure is associated with:

- Annoyance
- Disrupted sleep cycles
- Cardiovascular disorders
- Impaired cognitive development of children
- Worsened psychological and physical well-being ³⁵
- Seizures (audiogenic epilepsy) ³⁶

NO_x

Methane gas turbines pump harmful nitrogen oxides (mainly NO_2) into the air, which are known to cause:

- Cancer
- Asthma
- Respiratory diseases ^{4, 26}

Children are especially susceptible to respiratory conditions.

Finance

- Monopolizing goods for consumers leading to cost skyrocketing
 - Ram & Hard disk prices
- shift towards subscription based models and away from owning based logic

Sociopolitical Aspects

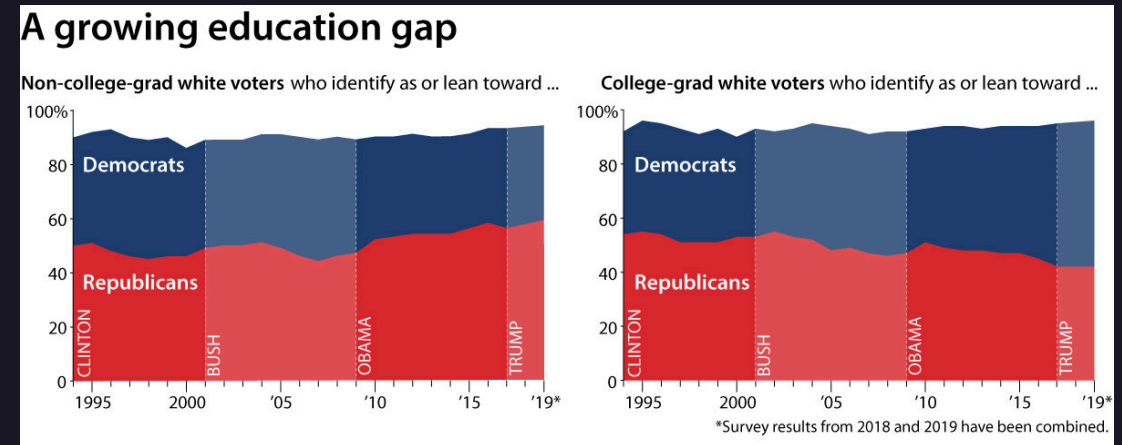
Job Creation

Stargate

- **100,000–200,000** construction and operations jobs ⁹
- **~25,000** onsite jobs ¹¹
- Large automated data centers often need *only dozens* of permanent staff once built
- May only employ **~57** ongoing workers, despite promises of thousands ¹²

Strategic Datacenter Placement 🐘

- Republican states often tend to be less educated
-
- Low backlash from 🐘 communities near sites
-



Ownership

Shift towards cloud computing

- Less burden on the user's device 🖥️
- Can sell less powerful devices, useless without cloud computing assistance ⁴⁰
- Cloud storage, cloud gaming, cloud movies, cloud music, cloud AI, cloud everything ☁️

Cloud services instigate subscription based models which degrade ownership.

[37]:

[38]:

